Date
Page

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Q. A positive integer n, a and b, return n^{th} magical number. Since the answer may be very large, return 10^9+7.

T1: n=1, a=2, b=3
output = 2

T2: n=4, a=2, b=3
output = 6

Algorithm:

Brute-Force → O(n) —— (X)
Not optimal

→ optimal:

1. Calculate LCM of a and b (numbers divisible by both
2. Search Range: smallest possible magical number (min(a,b)) to largest possible magical number (n* min(a,b))
3. Use binary search within this range
4. Count how many numbers upto middle number are divisible by a or b using Inclusion-Exclusion Principle.
5. If count < n, move the search to the right half.
6. Otherwise, move to the left half.

Date
Page

(7.) Repeat until search range reduces to one number.

(8.) The remaining number is the nth magical number.

For Test case - 2:

$lcm(2,3) = 6$

$low = 2$, $high = 2 \times 4 = 8$

$\therefore$ we search in $[2,8]$

Iteration: 1: $mid = (2+8)/2 = 5$

$count = \frac{5}{2} + \frac{5}{3} - \frac{5}{6} = 2+1-0 = 3$

Here, $count = 3 < n = 4$ (need larger numbers)

update: $low = mid + 1 = 6$

Iteration: 2 → $mid = (6+8)/2 = 7$

$count = \frac{7}{2} + \frac{7}{3} - \frac{7}{6} = 3+2-1 = 4$

$count >= 4$ → move left

$high = mid = 7$

Iteration 3: $mid = (6+7)/2 = 6$

$count = \frac{6}{2} + \frac{6}{3} - \frac{6}{6} = 3+2-1 = 4$

$count >= n$ → move left

$high = mid = 6$

Iteration 4: $low = high = 6$ → stop.

$\therefore$ th magical no = 6